Preliminary Exam 2025-26

Std : X
Time : 2 Hours

Sub. : Maths (II)

Marks : 40

Q.1) A) Four alternative answers are given for every subquestions. Select the correct alternative and write the alphabet of that answer. 4

1) From the following points ..... point lies to the right side of the origin on X-axis.

(a) (-2, 0) (b) (0, 2) (c) (2, 3) (d) (2, 0)

2) $\triangle ABC$ and $\triangle DEF$ are equilateral triangles. $A(\triangle ABC) : A(\triangle DEF) = 1 : 2$ If $AB = 4$ then what is the length of DE?

(a) $2\sqrt{2}$ (b) 4 (c) 8 (d) $4\sqrt{2}$

3) Out of the following which is the pythagorean triplet?

(a) (1, 5, 10) (b) (3, 4, 5) (c) (2, 2, 2) (d) (5, 5, 2)

4) The value of $9\cot^2\theta - 9\operatorname{cosec}^2\theta$ is .....

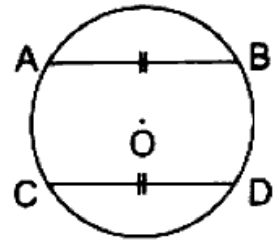
(a) 9 (b) -9 (c) 1 (d) -1

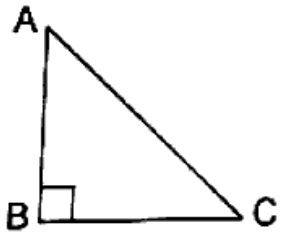
B) Solve the following questions. 4

1) Find the coordinate of the midpoint of the segment joining the point (22, 20) and (0, 16)

2) The volume of a frustum-shaped bucket is 10780 cm^3. What is the volume in litres?

3) Chord AB and Chord CD of a circle with centre O are congruent. If m (arc AB) = 120°, then find the m (arc CD).

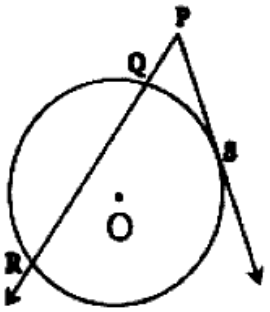


4) 

In $\triangle ABC$, $\angle ABC = 90°$, $\angle BAC = \angle BCA = 45°$ If $AC = 9\sqrt{2}$ then find the value of AB

Q.2) A) Complete the following activities and rewrite them. (Any two) 4

1) In the given figure, O is the centre of the circle, Seg PS is a tangent segment and S is the point of contact. Line PR is a secant. If PQ = 3.6, QR = 6.4, find PS.



(P.T.O)

https://www.maharashtrastudy.com

Solution :

$PS^2 = PQ \times$ [ ]

(Tangent secant segment theorem)

$= PQ \times (PQ +$ [ ] $)$

$= 3.6 \times (3.6 + 6.4)$

$PS^2 = 3.6 \times$ [ ]

$PS =$ [ ] ..... By taking square root.

2) If $\sec\theta = \frac{25}{7}$, find the value of $\tan\theta$.

$1 + \tan^2\theta = \sec^2\theta$

$1 + \tan^2\theta = \left(\frac{25}{7}\right)^{\square}$

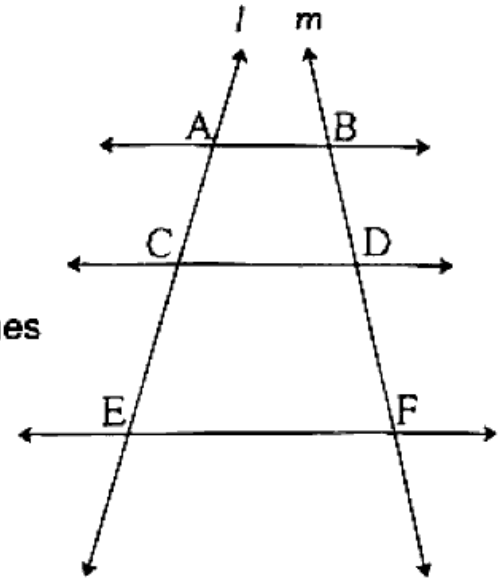
$\therefore \tan^2\theta = \frac{625}{49} - \square$

$= \frac{625 - 49}{49}$

$\tan^2\theta = \frac{\square}{49}$

$\therefore \tan\theta = \frac{\square}{7}$ .... By taking square root.

3) In the given figure, line AB ∥ line CD ∥ line EF
line l and line m are its transversals.
If AC = 6, CE = 9, BD = 8,
then complete the following activity to find DF.



Activity :

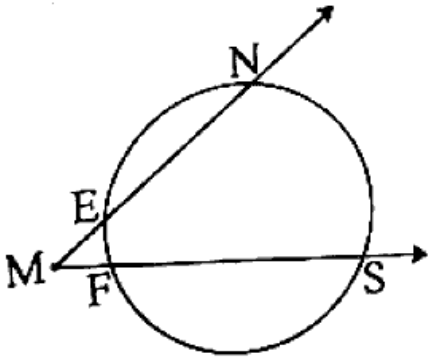
$\frac{AC}{\square} = \frac{\square}{DF}$ - Property of 3 parallel lines and their transversals.

$\therefore \frac{6}{9} = \frac{\square}{DF}$

$\therefore DF = \square$

B) Solve any four subquestions from the following. 8

1) In the figure,
m (arc NS) = 125°
m (arc EF) = 37°
find the measure
∠NMS



(P.T.O)

https://www.maharashtrastudy.com

2) Find the ratio in which T (-1, 6) divides the line segment joining the points P (-3, 10) and Q (6, -8)

3) Construct a tangent to a circle with centre O and radium 2.8 cm. at a point Q on it.

4) Measure of an arc of a circle is 80° and it's radius is 18 cm. Find the length of the arc. ($\pi = 3.14$)

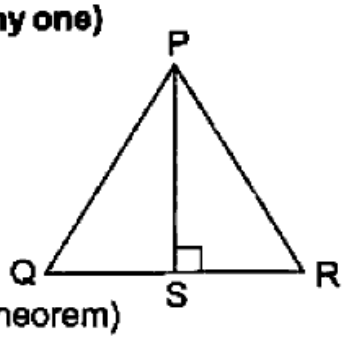
5) Prove that $\cos^2\theta \ (1 + \tan^2\theta) = 1$

Q.3) A) Complete the following activity and rewrite it. (Any one) 3

1) In $\triangle$ PQR, Seg PS $\perp$ side QR

then complete the activity to prove

$PQ^2 + RS^2 = PR^2 + QS^2$



Activity : In $\triangle$ PSQ, $\angle PSQ = 90°$

$\therefore \ PS^2 + QS^2 = PQ^2$ ..... (Pythagoras Theorem)

$\therefore \ PS^2 = PQ^2 -$ [ ] ..... (1)

Similarly,

In $\triangle$ PSR, $\angle PSR = 90°$

$\therefore \ PS^2 +$ [ ] $= PR^2$ ..... (Pythagoras Theorem)

$\therefore \ PS^2 = PR^2 -$ [ ] ..... (2)

$\therefore$ from (1) and (2)

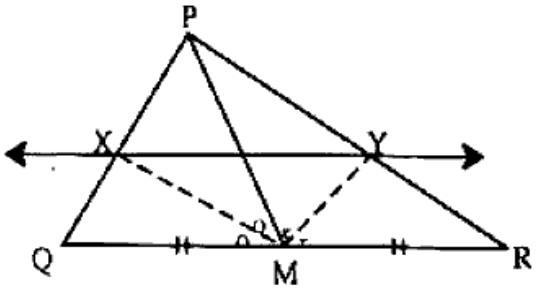
$PQ^2 -$ [ ] = [ ] $- RS^2$

$\therefore \ PQ^2 +$ [ ] $= PR^2 + QS^2$

2) In $\triangle$ PQR, seg PM is a median.

Angle bisectors of $\angle PMQ$ and $\angle PMR$ intersect side PQ and side PR in point X and Y respectively.

Prove that XY || QR.



Complete the proof by filling in the boxes. https://www.maharashtrastudy.com

Solution : In $\triangle$PMQ, Ray MX is the bisector of $\angle PMQ$

$\therefore \ \frac{MP}{MQ} = \frac{\square}{\square}$ .... (1) (Theorem of angle bisector)

Similarly in $\triangle$PMR, Ray MY is the bisector of $\angle PMR$

$\therefore \ \frac{MP}{MR} = \frac{\square}{\square}$ .... (2) (Theorem of angle bisector)

But $\frac{MP}{MQ} = \frac{MP}{MR}$ .... (3) (As M is the midpoint of QR)

Hence MQ = MR

(P.T.O)

https://www.maharashtrastudy.com

$\therefore \frac{PX}{\square} = \frac{\square}{YR}$ ..... from (1), (2) and (3)

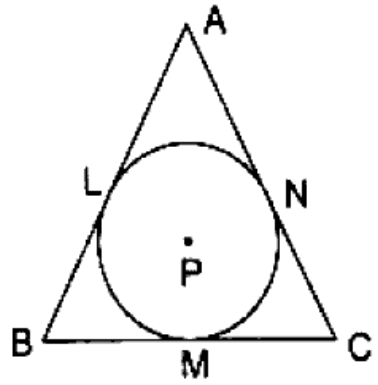
$\therefore$ XY ∥ QR ..... (Converse of BPT)

B) Solve the following subquestions. (Any two) 6

1) Prove that opposite angles of a cyclic quadrilateral are supplementary.
2) Draw a circle with centre P and radius 3.5 cm. Draw a minor arc AB of measure 120° in it. Draw tangents to the circle throught points A and B.
3) Show that the point A(1, 2), B (1,6), C ($1 + 2\sqrt{3}$, 4) are the vertices of an equilateral triangle.
4) The radius of a metallic sphere is 9 cm. It was melted to make a wire of diameter 4 mm. Find the length of the wire. ($\pi = 3.14$)

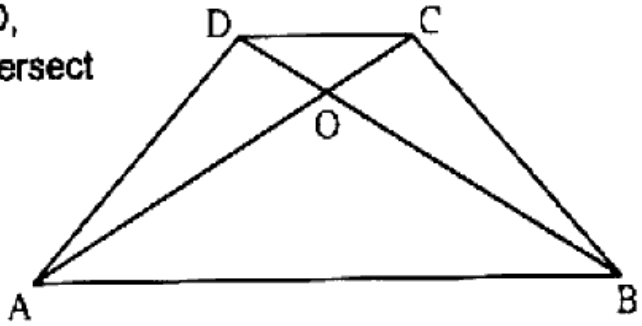
Q. 4) Attempt any two subquestions from the following. 8

1) Radius of circular base of an ear of corn is 6.6 cm and its length is 11.2 cm. If on an average 1 sq.cm. area contains 2 corn kernels, find the total number of Kernels on a corn.
2) A bird is sitting on the top of a 80m high tree. From a point on the ground, the angle of elevation of the bird is 45°. The bird flies away horizontally in such a way that it remains at a constant height from the ground. After 2 second, the angle of elevation of the bird observed from the same point is 30°. Find the speed of the bird. ($\sqrt{3} = 1.732$)
3) A circle with centre P is inscribed in the $\triangle ABC$. Side AB, side BC, side AC touch the circle at point L, M and N respectively. Radius of the circle is r. Prove that -
$A(\triangle ABC) = \frac{1}{2}(AB + BC + AC) \times r$



Q. 5) Solve the following subquestions. (Any one) 3

1) Draw two concentric circles having radii 3 cm and 5 cm. respectively. Take any point A on larger circle. Draw tangents to smaller circle from point A. Measure and write the length of tangent segment.
2) In trapezium ABCD, seg AB ∥ seg CD, AB = 3 CD. Diagonals AC and BD intersect each other in point O.
   a) Prove that :
   $\triangle AOB \sim \triangle COD$
   b) Hence prove :
   AO = 3 CO and OB = 3 OD



https://www.maharashtrastudy.com